B5 Muon g-2 framework continuation v0.3 prep — Feynman → K-type explicit translation roadmap

Lyra

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B5 Muon g-2 framework continuation v0.3 prep

Status entering v0.3

Closed (T2368 + T2374, Monday): - v0.1: K-type ↔ A_n structural mapping at I-tier (A_2 ↔ λ_W(3,3) = 42 = C_2·g; A_3 = HVP ↔ λ_W(2,2) = 24 = χ_K3; A_4 ↔ N_max−n_C−1 = 131 spectral gap) - v0.2: null-check survives Cal 6-failure-mode audit at I-tier (4 modes clear, 2 partial concerns identified) - Cal Task #23 verdict: STRONG I-tier across T2071+T2073+T2084+T2122 D-tier numerical match retained, mechanism-mapping at I-tier - Paper-text relabels applied per Keeper governance to Paper #106 v0.4 + section 5 + Paper #111 line 18 + Paper #115 v0.5_PRE line 128

OPEN for v0.3 (multi-day per Keeper Task #181): - Explicit Feynman-diagram → Wallach K-type translation - A_6 = 4500 prediction verification (decade-scale, Kinoshita group) - HVP + HLbL mechanism derivation from explicit Bergman Dirac propagators

v0.3 framework structure (substantive content scope ~3-5 days)

Phase A: Feynman-diagram → K-type structural map (~1-2 days)

For each known QED loop order n = 1, 2, 3, 4, 5: - Standard QED diagram set for muon g-2 contribution at order α^n - Identification of Wallach K-type (m_1, m_2) corresponding to each diagram class - Verification that the K-type Dirac eigenvalue gives the correct A_n coefficient

Specific mappings to verify:

Loop n	A_n	Predicted K-type	Diagram class
1	$\alpha/(2\pi)$	trivial vacuum bubble	Schwinger 1-loop
2	42/55	(3,3) λ_W = C_2·g = 42	2-loop vertex correction
3	24	(2,2) λ_W = χ_K3 = 24	3-loop with photon self-energy
4	131	spectral gap N_max−n_C−1	4-loop full QED set
5	750	higher K-type combination	5-loop Kinoshita

Phase B: HVP + HLbL mechanism (~1 day)

- HVP recurrence at λ_W(2,2) = χ_K3 = 24 is the same K-type as A_3
- This is internal Type C convergence (Lyra observation in T2368)
- Mechanism: vacuum-polarization diagrams share Bergman Dirac propagator with 3-loop QED at K-type (2,2)
- HLbL coefficient N_c^2·n_C = 45 corresponds to T2358 Type C 4-way (M_24 EOT moonshine)

Phase C: A_6 prediction verification (decade-scale)

- Prediction: A_6 = $rank^2 \cdot N\_c^2 \cdot n\_C^3$ = 4500 (T2122)
- Kinoshita group computing 6-loop QED in 2030s timeframe
- BST predicts falsifier: A_6 outside [3000, 6000] breaks the closed-form pattern
- Per Cal coincidence-filter discipline: this is the strongest single experimental test of B5 mechanism (forward verification at long horizon)

Calibration discipline maintained per Cal #23

Throughout v0.3 substantive drafting: - A_n numerical matches stay D-tier on the precision (per T2071+T2073+T2084+T2122) - K-type mechanism mapping stays I-tier (per Cal Task #23 verdict) - NO claim of "BST closes muon g-2" without explicit qualifier - All references to muon g-2 in papers carry Cal #23 qualifier per Monday relabel

v0.3 deliverable plan

Phase	Owner	Scope
A: Feynman→K-type map	Lyra	1-2 days, toy-verified per diagram class
B: HVP+HLbL mechanism	Lyra	1 day, structural identification
C: A_6 forward verification framing	Lyra	0.5 day, paper-grade framing for Paper #118 v0.3 Section 8.2 NEW
Cross-CI gate-pass	Cal	~1 wk after v0.3 lands

Strategic positioning

v0.3 completion would close Paper #118 v0.2 Section 9 named open item "Feynman-diagram → Wallach K-type explicit translation (~multi-week)" at the structural-mechanism level. Numerical-precision-mechanism closure remains open until A_6 verification (decade-scale).

Per Keeper's 4-paper spring coherence: - Paper #118 v0.2 ← Section 9 named open partially closed by B5 v0.3 - B5 muon g-2 mechanism becomes a Paper #118 v0.3 Section 8 substantive expansion

Honest scoping for v0.3 future work

Open after v0.3: - A_6 prediction verification (decade-scale, Kinoshita group) - Full per-flavor K-type SM fermion assignment (multi-month, Paper #118 v0.2 Section 9 named open) - Heat-kernel Tr(e^{-tD^2_full}) evaluation at non-origin Hua coords (multi-week, LAG-1 Session 9 v0.2) - Index theorem connecting muon g-2 to Atiyah-Singer (multi-month, LAG-1 Session 10)

Per Cal External_Survivability_Checklist: NOT a positive claim about B5 D-tier mechanism. Pre-staged framework for multi-day substantive derivation. Each phase tier-labeled honestly.

Filing notes

Status: v0.3 prep filed per Casey "do all" directive 2026-05-18 PM. Multi-day substantive drafting queued.

Owner: Lyra (B5 primary, Keeper Task #181).

Next pull: when Lyra has 1-2 day window, Phase A (Feynman→K-type map) is the substantive opening.

— Lyra, B5 v0.3 framework continuation prep filed per Casey "do all" directive 2026-05-18 PM.